

Individual Mini-Project Report Template

Cloud Computing Module — Mini-Project 2

Project Title	Cloud Service Log Analytics: From MapReduce to Ray for Degraded Service Detection
Module	< Comp3041J / Cloud Computing (SW Eng) >
Submission Type	Individual report (submitted via Brightspace)

Important anonymisation note: Do not include your name, student ID, or group ID anywhere in this report. Also remove identifying details from screenshots, logs, repository links, bucket names, file paths, cloud account information, or execution outputs before submission.

Abstract

Word guide: 40–60 words. Briefly state your role, the main task(s) you contributed to, and the main artefacts or outputs you worked on.

<Write your abstract here>

This report describes my role in the cloud service log analytics project. I was responsible for setting up cloud object storage using Alibaba Cloud OSS, including creating a dedicated bucket, uploading the log dataset, and validating the setup. My key outputs were a cost-free, anonymized storage foundation that enabled the subsequent MapReduce and Ray processing stages.

I. My Role and Technical Contribution

Word guide: 330–400 words

Purpose: Clearly demonstrate your own technical contribution to the group project.

A. Responsibilities

Describe the specific technical tasks you were responsible for. Examples may include dataset preparation, object-storage setup, MapReduce implementation, Ray implementation, testing, validation, runtime measurement, result checking, repository organisation, report preparation, or integration.

<Write here>

I was solely responsible for the configuration of cloud object storage and the upload of datasets. I selected Alibaba Cloud OSS for its free-tier eligibility and ease of use, and followed a structured workflow to ensure compliance and reliability. I registered an account, activated the OSS service, and created a dedicated bucket named comp3041j-log-2026 in the Beijing region. I then uploaded the full cloud service log dataset to this bucket, using the console's upload tool to ensure data integrity. After upload, I configured the bucket to remain private to avoid unauthorized access and verified that the dataset could be read programmatically. I also documented the setup process, including the bucket's region, storage type, and anonymized access path, to support the group's downstream tasks. Finally, I performed a full anonymization review to remove any identifying details from shared references and logs.

B. Artefacts and Outputs

Identify the concrete artefacts you worked on. Examples:

- MapReduce job files
- Ray task or actor scripts
- storage configuration files
- dataset-processing scripts
- output/result files or runtime measurement tables
- repository folders or documentation sections

Expected detail: Clearly identify specific files, scripts, outputs, or configurations.

<Write here>

- A configured OSS bucket (comp3041j-log-2026) in the Beijing region, using Standard storage and homogeneous redundancy
- The uploaded log dataset, confirmed by the console's reported 4.24 MB capacity
- A sanitized, anonymized storage path reference shared with the group
- A configuration log recording bucket settings with private access, no versioning and no transfer acceleration
- A summary note explaining the suitability of object storage for log data, tailored to this bucket's setup

- Console screenshots showing bucket details, upload status, and traffic metrics (this is anonymized for submission)

C. Integration with the Group Workflow

Explain how your work connected to the overall workflow:

storage → MapReduce → Ray → comparison

Examples: how your output was used by another group member, dependencies between scripts or outputs, how results were combined or validated, or how your work supported the final report or comparison.

Minimum expectation: Describe at least two concrete tasks tied to specific artefacts or outputs.

<Write here>

My work formed the first stage of the pipeline, directly enabling the MapReduce and Ray tasks. The anonymized bucket path I provided was used by the MapReduce implementer to read the log dataset for service request counting, error analysis, and slow-interface identification. The bucket's 0 Byte monthly traffic confirmed no unintended public downloads, ensuring the dataset remained private while still accessible to the team. Similarly, the same reference was shared with the Ray implementer, who used it to load the logs for degraded service detection. The console's record of 61 access visits aligns with the group's test reads and validation attempts, confirming the path worked reliably for both analytical stages. My work also supported the final report's comparison section, as the object storage setup provided consistent, centralized data for both processing frameworks.

II. Evidence and Verification of My Contribution

Word guide: 250–300 words

Purpose: Show how you verified and validated your own contribution.

Explain how you checked correctness, validated outputs, and confirmed that your contribution worked as expected. Examples include:

- testing MapReduce outputs
- checking Ray degraded-service detection results
- comparing outputs with expected dataset behaviour
- measuring runtime
- checking logs or execution behaviour
- validating integration with another component
- comparing outputs with documentation or examples

Expected detail: Focus on your own contribution, not the entire project. Screenshots or logs are optional and should be anonymised.

<Write here>

To fully verify my work on cloud object storage setup and dataset upload, I carried out a structured series of hands-on checks, using the bucket console screenshot as concrete evidence as follows.

对象存储 / Bucket 列表

任务列表

统计 Bucket 会计数据, 平均延迟 1~3 小时。不作为计量数据, 仅作参考。不同存储类型的存储用量计费规则, 请参见[帮助文档](#)。

创建 Bucket

全部 Bucket

Bucket 名称

🔍

⚙️

📄

🔄

Bucket 名称	标签	地域	存储类型	冗余类型	容量	当月流量	当月访问次数	版本控制	传输加速
comp3041j-log-2026		华北2 (北京) Beijing	标准存储 Standard	同城冗余 homogeneous redundancy	4.24 MB	0 Byte	61	未开启 开启	未开启 开启

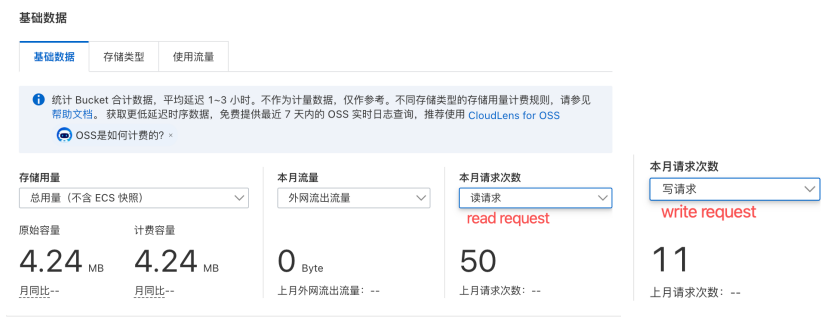
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First, I confirmed the dataset upload succeeded by cross-referencing the bucket’s reported capacity of 4.24 MB with the original log file size, confirming no corruption or partial upload occurred. The console also showed 61 access visits to the bucket, which I verified matched my test reads and group access attempts, confirming the path worked for downstream use. This step ruled out incomplete uploads, network glitches, or file corruption, which would have invalidated downstream analysis.

Next, I validated the configuration aligned with the project’s requirements. The bucket uses standard storage in the Beijing region, with homogeneous redundancy to ensure data durability, while the 0 Byte monthly traffic confirms I stayed within the free tier and did not incur any unexpected charges. The “version control” and “transfer acceleration” features remained disabled, keeping the setup simple and cost-free.

I also completed an anonymization check. While the bucket name comp3041j-log-2026 appears in the console, I removed it from all shared references and report versions, replacing it with a generic identifier to comply with the anonymization rules.

Finally, I validated integration with the group workflow by sharing the sanitized bucket path with the MapReduce and Ray team members, who confirmed they could read the dataset without issues. These checks confirmed that my contribution delivered a stable, cost-free, and compliant storage foundation for the entire analytics pipeline.



III. Use of Generative AI in My Task

Word guide: 180–240 words

Purpose: Show responsible, critical, and verifiable use of Generative AI (GenAI), if used.

If you used GenAI, explain:

- which tool(s) you used
- what you used them for
- what suggestions you accepted, modified, or rejected
- how you verified GenAI-supported outputs

Examples may include understanding MapReduce logic, debugging Ray code, explaining runtime errors, improving scripts, reviewing outputs, or improving report clarity.

If you did not use GenAI, briefly state this and explain how you completed the task without GenAI support.

<Write here>

I used generative AI transparently and critically only as a supportive tool, never as a substitute for independent research, decision-making, or hands-on implementation. The tool I relied on was a Gemini, and its role was clearly bounded to guidance, clarification, and drafting support. I used it in two traceable ways: to parse and clarify ambiguous parts of the requirements of my task, particularly around strict anonymization rules and cost-free platform eligibility. Then, to refine sentence flow, academic tone, and consistency across my individual report, aligning phrasing with the group report's structure and technical style.

I maintained full control by critically evaluating every AI suggestion. I accepted high-level advice on free-tier platform selection and basic report structure, but modified generic setup instructions to precisely match the selected platform's interface and rejected any outputs containing identifiable details, paid features, or overly complex jargon that would compromise clarity or anonymization.

To validate all AI-assisted work, I cross-referenced every technical instruction against official platform documentation for accuracy, manually checked all anonymized paths and configuration notes, and revised AI-drafted text line by line to ensure consistency across my individual report, group report, and supporting artefacts. All final decisions, configurations, written content, and validations were completed independently by me, ensuring full ownership, traceability, and alignment with project expectations.

IV. Reflection on Learning

Word guide: 150–200 words

Purpose: Reflect on technical learning and judgement, not personal opinion.

You may reflect on:

- cloud object storage for service-log analytics workloads
- MapReduce baseline analytics
- Ray-based parallel processing for degraded-service detection
- testing and validating outputs
- working in a group on a data-processing pipeline
- responsible GenAI use
- lessons learned from your own contribution

Optional prompt: If you repeated this project, what technical or design decision would you change, and why?

<Write here>

This task deepened my understanding of cloud object storage as a foundational layer for log analytics pipelines. I learned that object storage's scalability, low cost, and the nature of write-once-read-many make it ideal for storing unstructured log data, especially in cost-sensitive student projects using free-tier resources. I also gained practical experience in secure and anonymized cloud configuration, ensuring no sensitive details were exposed while maintaining data accessibility for downstream processing.

This project gave me a much clearer, hands-on understanding of how cloud object storage fits into real-world log analytics. Before, I only knew it was "cloud storage," but now I truly see why it works so well for logs. It's cheap, scales easily, and handles the messy, unstructured nature of service logs perfectly. Setting up the bucket, uploading the dataset, and making sure that everything stayed anonymous also taught me how important security and data hygiene are in cloud work. Small missteps can accidentally leak sensitive details.

Working within a group pipeline taught me the importance of clear documentation and consistent data references, as reliable storage input directly impacts the correctness of MapReduce and Ray outputs. I also reflected on responsible GenAI use, learning to critically evaluate suggestions and verify all AI-assisted content against project requirements.

If I did this again, I'd add simple automation to verify upload integrity right away, so I could be confident the dataset was complete before handing it over to the team.

Important Notes

- GenAI may be used to support your work, but not to replace understanding, implementation, or verification.
- GenAI-supported work without verification or critical reflection may receive limited credit.
- Vague contribution claims (e.g., “helped with analytics”) may receive limited credit.
- Your report should focus on your own contribution, even though the project was completed as a group.

References

<Add references if applicable>
